Modern C++17 Smart Pointers & Memory Management • INSTRUCTOR ANSWER KEY

Course: C++ 17 • College: DPU • Student Name: _____

Q1. Which runtime construct is introduced in C++ 17?

```cpp
#include <iostream>
#include <memory>

struct Buffer {
    int id;
    Buffer(int i) : id(i) {}
};

std::unique_ptr<Buffer> make_buffer(int id) {
    auto buf = std::make_unique<Buffer>(id);
    return buf; // Move semantics applied
}
```

[CORRECT] A) Managed lightweight runtime scheduler

B) Static manual thread preemption

C) Unbuffered heap allocation

D) Deprecated bytecode interpreter

Key: A (Grounded in Slide 2) — Grounded in Slide 2: Unique Pointer & Move Semantics

Distractor Analysis (Common Mistakes):